Smart Traffic Signal Network using an Active Database

Course: Data Modelling
Project Type: Active Database Design and Simulation
Technologies: PostgreSQL, C/C++, libpq

1. Introduction

Urban traffic congestion is a major issue in modern cities. Fixed-time traffic signals do not adapt well to changing traffic conditions, which leads to longer waiting times and inefficient vehicle movement.

This project proposes a smart traffic signal network implemented using an **active database**. The system models the road network as a **directed graph** in which junctions are nodes and road segments are edges. Incoming traffic at junctions is represented through approaches, while signal control is handled through phases. The database automatically reacts to sensor updates using triggers and stored procedures.

2. Objectives

The objectives of the project are:

- model a connected traffic network in a relational database
- represent junction connectivity using directed road segments
- process live traffic data through simulated sensors
- implement active database behavior using PostgreSQL triggers
- dynamically adjust signal timings based on traffic density
- support corridor-based synchronization across junctions
- maintain traffic history for analysis

3. System Overview

The system consists of:

- a PostgreSQL database storing the traffic network and signal data
- a C/C++ simulation module that generates traffic input
- trigger-based active logic that updates queue and phase information
- a controller or simulator that reads the updated signal decisions

The database stores the network structure, receives sensor readings, updates traffic state, and recomputes signal timing automatically.

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4. Database Design

The redesigned schema separates **road network structure** from **signal control logic**.

- **Junctions** represent intersections
- **Road segments** represent directed roads between junctions
- **Approaches** represent incoming traffic streams at a junction
- **Movements** represent valid transitions from an incoming approach to an outgoing road segment
- **Signal phases** represent groups of traffic movements that can be active together
- **Sensors** monitor approach-level traffic
- **Sensor readings** act as event data for active database behavior
- **Corridors** represent ordered routes used for synchronized scheduling

This graph-based design is more realistic than the earlier lane-based model because it explicitly captures connectivity between roads and junctions.

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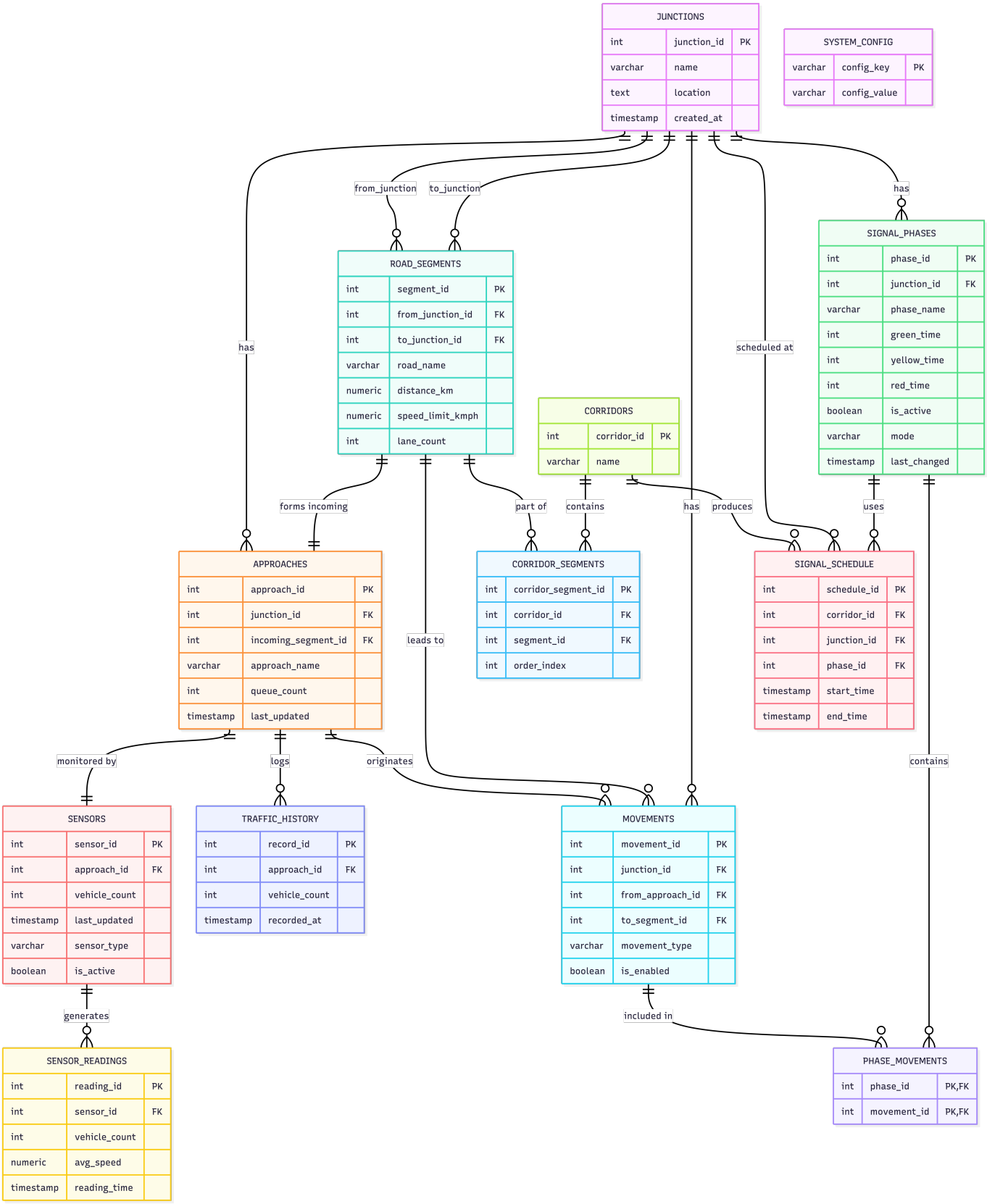
5. Database Schema Diagram

The following diagram presents the final relational schema of the smart traffic signal network. The design models the road network as a directed graph, where junctions act as nodes and road segments act as edges. It also shows the control layer of the system, including approaches, movements, signal phases, sensors, and corridor-based scheduling. This schema forms the foundation of the active database and supports both adaptive signal control and coordinated traffic management.



6. Active Database Behavior

The system follows an Event–Condition–Action style design.

Event: insertion of a new sensor reading
Condition: current queue level on an approach
Action: update queue count, log history, and recompute signal timing

Whenever a new reading is inserted:

1. the sensor state is updated
2. the approach queue count is updated
3. a traffic history record is inserted
4. the corresponding signal phase is recalculated

This allows the database to actively respond to traffic changes instead of working only as passive storage.

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7. Signal Control Logic

Signal timing is adjusted using configurable threshold values stored in the database.

- under low traffic, the system assigns a smaller dynamic green time
- under moderate traffic, the green time is increased proportionally
- under heavy traffic, the system switches to periodic mode for stability

This makes the traffic signal system adaptive to different traffic conditions.

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8. Corridor Synchronization

The system also supports smart corridor scheduling. A corridor is modeled as an ordered sequence of road segments. Using segment distance and average travel speed, the database computes phase timing offsets across junctions.

This enables coordinated signal timing, or a **green wave**, so that vehicles moving along an important route can pass through multiple junctions with fewer stops.

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9. Testing

The database was tested using SQL scripts covering:

- schema creation
- sample data loading
- sensor-driven updates
- dynamic phase adjustment under low, moderate, and heavy traffic
- manual phase override
- corridor schedule generation

The tests confirmed that triggers, stored procedures, and scheduling logic work correctly on the sample network.